

Sample gain and the study-only redundancy

Draft note for *Crossing the Validation Crisis*

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This note defines the redundancy $\omega = t \rho_e$ (test fraction times the loss correlation of two split predictors) and its study-only estimate $\hat{\omega}_k^{\text{study}}$, and derives the reference curve $G_K^{\text{test}} \approx K/(1 + (K - 1)\omega)$ that relates it to the variance-equivalent test sample gain. It replaces the former redundancy score $\hat{C}_{g,k}^{\text{study}} \hat{\rho}_{e,k}^{\text{study}} \bar{m}_k^{\text{study}}$, and explains why an absolute early-stopping threshold can be placed on the redundancy but not on the former score. Notation follows the paper; results taken from it are restated so that this note reads on its own, with a pointer to where they appear (section numbers of the current draft, appendix titles in full).

1 Setting and definitions

A study set of N observations is split K times by Monte-Carlo cross-validation (MCCV): each split a draws a test set $\mathcal{D}_a^{\text{te}}$ of size n_{te} uniformly without replacement, independently across splits, and trains the predictor $g_a = F_\lambda(\mathcal{D}_a^{\text{tr}}, \xi_a)$ on the remaining $n_{\text{tr}} = N - n_{\text{te}}$ observations. The test fraction is

$$t = \frac{n_{\text{te}}}{N}. \quad (1)$$

For an observation i held out by split a , let

$$d_{a,i} = \ell(g_a(x_i), y_i) - \mathcal{R}^*(g_a) \quad (2)$$

be its per-sample loss centred at the risk of the predictor that did not see it, so that $\mathbb{E}[d_{a,i} | g_a] = 0$, and let $\sigma_e^2 = \mathbb{E}[d_{a,i}^2]$ be the per-sample loss variance.

The benchmark-adjusted fold error of split a is $\delta_a = R_{\mathcal{D}_a^{\text{te}}}(g_a) - \hat{R}^*(g_a)$, and the K -split estimation error is $\Delta_K = \frac{1}{K} \sum_{a=1}^K \delta_a$. When the evaluation noise of the benchmark estimate $\hat{R}^*$ is negligible,

$$\delta_a = \frac{1}{n_{\text{te}}} \sum_{i \in \mathcal{D}_a^{\text{te}}} d_{a,i}. \quad (3)$$

This is the evaluation-noise approximation of the paper (Section 3, paragraph “Evaluation-noise approximation”; Appendix B, “Measuring and approximating the evaluation noise”). With $\sigma^{2,\text{te}} = \text{Var}(\delta_a)$ and $\tau^{\text{te}} = \text{Cov}(\delta_a, \delta_b)$ for $a \neq b$, the exchangeability of the splits gives

$$\text{Var}(\Delta_K) = \frac{1}{K} \sigma^{2,\text{te}} + \frac{K-1}{K} \tau^{\text{te}}. \quad (4)$$

(paper, Section 3, variance of Δ_K ; Appendix C, “Benchmark-adjusted fold errors”).

2 Step 1: the gain depends on the fold-error correlation only

The variance-equivalent test sample gain $G_K^{\text{test}} = N_K^{\text{equiv}}/n_{\text{te}}$ is the factor by which a single hold-out test set must be enlarged to match the variance of Δ_K . When the single-split variance scales as $1/m$ with the test size m , it equals the variance ratio

$$G_K^{\text{test}} \approx \frac{\text{Var}(\delta_a)}{\text{Var}(\Delta_K)} = \frac{K}{1 + (K-1)\rho_\delta}, \quad \rho_\delta = \frac{\tau^{\text{te}}}{\sigma^{2,\text{te}}} = \text{Corr}(\delta_a, \delta_b), \quad a \neq b, \quad (5)$$

where the second equality follows from (4). This is the intraclass-correlation form of the paper (Appendix D.1, “Benchmark-aware estimation of sample gain”); the gain itself is defined in Section 3 (Definitions “Variance-Equivalent Test Size” and “Test Sample Gain”). Up to the $1/m$ scaling, this step is exact: everything the gain depends on is in the fold-error correlation ρ_δ .

3 Step 2: decomposing the fold-error covariance

Single split. Given g_a , the held-out observations are i.i.d. and the $d_{a,i}$ are centred, so the cross terms within a split vanish and

$$\sigma^{2,\text{te}} = \frac{\sigma_e^2}{n_{\text{te}}}. \quad (6)$$

Two splits. For $a \neq b$, let $I_{ab} = \mathcal{D}_a^{\text{te}} \cap \mathcal{D}_b^{\text{te}}$ be the observations held out by both. Expanding the product of the two averages in (3) and separating the shared observations from the pairs of distinct observations,

$$\tau^{\text{te}} = \frac{1}{n_{\text{te}}^2} \left(\underbrace{\mathbb{E} \left[\sum_{i \in I_{ab}} d_{a,i} d_{b,i} \right]}_{\text{shared observations}} + \underbrace{\mathbb{E} \left[\sum_{i \in \mathcal{D}_a^{\text{te}}} \sum_{j \in \mathcal{D}_b^{\text{te}}, j \neq i} d_{a,i} d_{b,j} \right]}_{\text{train/test coupling}} \right). \quad (7)$$

Shared observations. Two independent uniform test sets share on average $\mathbb{E}|I_{ab}| = n_{\text{te}}^2/N = t n_{\text{te}}$ observations. On each of them the losses of the two split predictors covary by $\rho_e \sigma_e^2$, where

$$\rho_e = \frac{\mathbb{E}[d_{a,i} d_{b,i} \mid i \in I_{ab}]}{\sigma_e^2} \quad (8)$$

is the correlation of the per-sample losses of two split predictors on an observation that neither was trained on. The first term of (7) is therefore $t n_{\text{te}} \rho_e \sigma_e^2 / n_{\text{te}}^2$.

Train/test coupling. For $i \neq j$, $d_{a,i}$ and $d_{b,j}$ are not independent: i may belong to the training set of g_b , j to that of g_a , and g_a and g_b share most of their training data. This dependence is intrinsic to cross-validation and is part of τ^{te} , as the paper states when it introduces τ^{te} (Section 3; Appendix C, law of total covariance). Writing $\gamma = \mathbb{E}[d_{a,i} d_{b,j}]$ for the average covariance of such a pair, the second term is approximately γ , since there are about n_{te}^2 such pairs.

Result. Dividing by $\sigma^{2,\text{te}} = \sigma_e^2/n_{\text{te}}$,

$$\boxed{\rho_\delta = t \rho_e + \kappa, \quad \kappa = \frac{n_{\text{te}} \gamma}{\sigma_e^2}.} \quad (9)$$

The loss correlation ρ_e measures how similarly two split predictors err on the same observation. The factor t is the share of a test set made of observations that another split also tests. The coupling contribution κ can have either sign.

4 Step 3: the redundancy and the reference curve

Substituting (9) into (5),

$$G_K^{\text{test}} \approx \frac{K}{1 + (K-1)(t \rho_e + \kappa)}. \quad (10)$$

We call

$$\omega = t \rho_e \quad (11)$$

the *redundancy*: the part of the fold-error correlation carried by observations that two splits both test. With $\kappa = 0$, (10) is the reference curve $K/(1 + (K - 1)\omega)$. Two limits can be read directly. For $\rho_e \rightarrow 0$, split predictors err independently and $G_K^{\text{test}} \rightarrow K$. For $\rho_e \rightarrow 1$, all split predictors err alike and $G_K^{\text{test}} \rightarrow K/(1 + (K - 1)t)$, whose limit as $K \rightarrow \infty$ is $1/t = N/n_{\text{te}}$: cross-validation then averages over the whole study set and no more.

Study-only estimate of the redundancy. After the first k splits of a single run, let I_{ab} be the observations held out by both splits $a < b$ and $e_{a,i} = \ell(g_a(x_i), y_i)$ the per-sample loss, computed with the loss of the metric whose gain is targeted. The pair-averaged loss covariance, loss variance and their ratio are

$$\widehat{C}_{e,k}^{\text{study}} = \binom{k}{2}^{-1} \sum_{a < b} \widehat{\text{Cov}}_{i \in I_{ab}}(e_{a,i}, e_{b,i}), \quad (12)$$

$$\widehat{V}_{e,k}^{\text{study}} = \binom{k}{2}^{-1} \sum_{a < b} \frac{1}{2} (\widehat{\text{Var}}_{i \in I_{ab}}(e_{a,i}) + \widehat{\text{Var}}_{i \in I_{ab}}(e_{b,i})), \quad (13)$$

$$\widehat{\rho}_{e,k}^{\text{study}} = \widehat{C}_{e,k}^{\text{study}} / \widehat{V}_{e,k}^{\text{study}}, \quad (14)$$

(these are the loss covariance, variance and correlation of the paper, Appendix D.3, “Study-only redundancy for early-stopping cross-validation”), and the redundancy after k splits is

$$\widehat{\omega}_k^{\text{study}} = t \widehat{\rho}_{e,k}^{\text{study}}. \quad (15)$$

It is dimensionless and lies between $-t$ and t . It can be read after $k = 3$ or $k = 5$ splits: configuration-level relations are the same, while the spread of a single run’s reading roughly halves from $k = 3$ to $k = 5$. The empirical overlap $\bar{m}_k^{\text{study}}/n_{\text{te}}$, with $\bar{m}_k^{\text{study}}$ the mean of $|I_{ab}|$ over pairs, may replace t : it has the same expectation.

5 Relation to the former redundancy score

The former redundancy score of the paper (Appendix D.3, boxed equation; Section 5.1, paragraph “Measuring how sample gains settle in”) is $\widehat{s}_k = \widehat{C}_{g,k}^{\text{study}} \widehat{\rho}_{e,k}^{\text{study}} \bar{m}_k^{\text{study}}$, where $\widehat{C}_{g,k}^{\text{study}}$ is the pair-averaged covariance of the two split predictions on I_{ab} . Since $\mathbb{E} \bar{m}_k^{\text{study}} = t n_{\text{te}}$,

$$\widehat{s}_k \approx \widehat{C}_{g,k}^{\text{study}} n_{\text{te}} \widehat{\omega}_k^{\text{study}}. \quad (16)$$

The former score thus contains the redundancy, multiplied by the prediction covariance, which carries the squared units of the target and the strength of the signal, and by the test-set size, which grows with the study. Both factors vary across settings independently of the gain. The redundancy is dimensionless and bounded by t , which is why an absolute early-stopping threshold can be placed on it. On the prospective holdout of the paper (216 configurations, 50 seeds) with gains on the mean squared error, the former score is not informative (correlation with $\log G_{20}^{\text{test}}$ of +0.11 after 3 splits), while the redundancy is (−0.60 after 3 splits, −0.67 after 5).

6 Assumptions and scope

1. The evaluation noise of the benchmark estimate is negligible and the single-split variance scales as $1/m$ with the test size.

2. $\hat{\rho}_{e,k}^{\text{study}}$, computed on I_{ab} with sample centring, estimates ρ_e with a bias of order $1/|I_{ab}|$; the held-out folds must therefore contain a few observations.
3. The overlap $\mathbb{E}|I_{ab}| = t n_{\text{te}}$ holds for MCCV. For repeated K -fold, splits of the same repeat have disjoint test sets and the factor changes accordingly.
4. The reference curve omits the coupling term κ .

7 Empirical checks on simulated data

1. The study-only $\hat{\rho}_{e,k}^{\text{study}}$ matches, within 1% across all simulated designs, the population ρ_e of the same split predictors measured on the benchmarking set.
2. Varying the test fraction with everything else fixed, the fitted factor ρ_δ/ρ_e follows t (Table 1); fits with an intercept give intercepts within ± 0.01 .
3. Regression configurations and the **PCam** networks lie on the reference curve (**PCam**: median $\log(G_{20}^{\text{test}}/\text{curve})$ between -0.02 and $+0.04$). Simulated classifiers fall 10–50% below it at training sizes 100–300, that is $\kappa > 0$ there, and the gap closes by $n_{\text{tr}} \approx 800$.

Table 1: Fitted factor ρ_δ/ρ_e by test fraction (MCCV, $K = 20$, 50 seeds, training sizes 300 and 1000).

test fraction t	classification (accuracy)	regression (MSE)
0.1	0.10	0.14
0.2	0.21	0.20
0.3	0.32	0.36
0.5	0.48	0.46